
Revenue Forecasting for OTB Planning

From Data Engineering to 12-Month Forecast

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What is OTB Forecasting?

Open-to-Buy

Purchasing budget mechanism

That answers: *How much budget do I have left to spend on inventory, and where should I allocate it?*

Forecast Output Shapes

How much inventory to commit

When to commit it

At what cost

Accuracy matters. But direction of error matters more.

OTB at Ounass

THE BUSINESS

- Luxury e-commerce, UAE and KSA
- Low SKU volume, high AOV
- Event-driven demand: Ramadan, Eid, DSF

WHAT THIS MEANS FOR FORECASTING

- Importance of precision in peak period capture
- Monthly budget cadence: daily/weekly modeling aggregated up
- Market-specific patterns require market-aware features

DATA ENGINEERING

Structuring Signals

Three structural decisions shaped everything downstream.

Population Separation

11

Business Accounts Identified

6×

Higher Avg Revenue vs Consumers

CLASSIFICATION RULE

- $\text{qty} > 10 = \text{bulk order}$
- $\text{bulk orders} > 2 \text{ AND } \text{bulk freq} > 50\% = \text{B2B}$
(conservative take)
- Customer-level, not transaction-level

MODELING DECISION

- Too sparse for standalone model (11 accounts)
- Retained as `bulk_order_count` covariate per period
- Both populations contribute to OTB revenue target

Mixing populations corrupts both signals

Cancellation Treatment

18%

Zero-Revenue Rows

- Only 1 negative revenue row in 10M+ records, returns not captured
- Zero revenue = cancelled order (qty recorded, transaction incomplete)
- Cancellation count retained as per-period covariate, frequent cancellations in a category $\times$ market pair can be a demand signal

GCC-Specific Feature Engineering

Ramadan Weight

- Exponential curve $0 \rightarrow 1$ across the month
- Captures slump-to-peak trajectory
- Early: introspection, low consumption
- Late: Eid prep, gift buying, surge

Market-Specific Holidays

- UAE: National Day Dec 2, DSF Dec 6–Jan 12
- KSA: National Day Sep 23, Founding Day Feb 22
- Verified by aligning revenue spikes to dates
- Cross-market: New Year, Christmas (expat effect)

Forward-looking

- All event dates known in advance
- Full 12-month feature set at forecast time
- Zero leakage risk on forecast horizon

Seasonality Confirmed

PROPHET DECOMPOSITION

- Strong linear trend: 12M → 40M daily revenue over 3 years
- March peak dominant across both markets (Ramadan + Eid al-Fitr)
- Summer trough Jul–Sep: GCC heat and travel season
- UAE: distributed seasonality, DSF recovery Nov–Dec
- KSA: concentrated around Ramadan, flat otherwise

KEY INSIGHT

Market A and Market B have structurally different seasonal profiles.

Justifies market-aware modeling and separate holiday features.

Why Prophet?

Decomposition

Explicit trend + seasonality
+ holiday components

Multiplicative

Seasonal swings scale with
business growth

Regressors

GCC calendar features
added as explicit additives

Structure of the model matches structure of the data.

Baseline vs Regressor Model

74%

MAPE Reduction from Regressors

Model	MAPE	MAE (AED/day)	Bias
Vanilla	17.03%	4,131,943	+0.034
With Regressors	4.39%	1,088,790	-0.014

- MAPE: avg % error per period: scale-free, comparable across series
- MAE: error in AED: operationally meaningful for planners
- Bias: direction of error: positive = over-forecast, negative = under-forecast

GCC calendar features drive precision. Trend and seasonality shape alone are not enough.

Category × Market Breakdown

Series	MAPE	Bias
Category_A × Market_A	7.16%	-0.039
Category_A × Market_B	7.68%	+0.058
Category_B × Market_A	9.47%	-0.020
Category_B × Market_B	23.20%	+0.219
Category_C × Market_A	18.61%	+0.117
Category_C × Market_B	11.40%	-0.006
Category_D × Market_A	5.82%	+0.016
Category_D × Market_B	7.65%	+0.028
Category_F × Market_A	8.84%	-0.052
Category_F × Market_B	9.25%	+0.034

* Sparse series: high bias driven by low volume.

Alternative Considered: LightGBM

THE HYPOTHESIS

- Prophet models are isolated, so each sees one series
- LightGBM: all series stacked, category x market as features
- Can learn co-variation across series during shared events

THE RESULT

Model	MAPE	Bias
LightGBM	7.92%	-0.081
Prophet	4.39%	-0.014

Diagnosis: lag_365 cannot represent business growth or trend. Leads to systematic under-forecast

Next steps: explicit trend feature (YoY growth rate), early stopping

FORECAST QUALITY

Bias is a business risk, not just a metric.

In luxury OTB, over-forecast and under-forecast carry structurally different costs.

Asymmetric Cost of Error

-0.014

OVER-FORECAST

- Excess inventory committed
- Capital tied in unsold stock
- Markdown pressure on luxury AOV
- Warehouse cost without revenue return

UNDER-FORECAST

- Stockout: lost luxury sale
- Emergency reorder: higher unit cost
- Lead times can be long for luxury brands
- Customer lost, not just transaction lost

Model bias of -0.014 is slight under-forecast.. the safer direction, but requires monitoring at the series level.

Key Takeaways

- 1 Population separation and cancellation treatment were prerequisites
- 2 GCC-specific features (Ramadan curve, market holidays) serve as important signals
- 3 Prophet per-series: 4.39% MAPE, -0.014 bias, slight under-forecast, the safer direction for luxury OTB
- 4 LightGBM is a viable alternative once an explicit trend feature inherent time-indexing limitation